

DHCP server configuration for VLAN 10 (computer laboratory room/devices):

- The first IP address will be allocated for default gateway (192.168.10.1)
- Given the subnet mask of CIDR /27, subnet mask is 255.255.255.224
- Maximum number of users will be 29 hosts (minus the gateway IP address)

1st subnet:

- Network: 192.168.10.0/27
- Hosts: 192.168.10.1 - 192.168.10.30 (/27)
- Broadcast: 192.168.10.31/27

DHCP_SERVER

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: COMLAB_DHCP

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

Start IP Address: 192 168 10 2

Subnet Mask: 255 255 255 224

Maximum Number of Users: 29

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Buttons: Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

☐ Top

DHCP server configuration for VLAN 20 (HR Department):

- The first IP address will be allocated for default gateway (192.168.10.33)
- Given the subnet mask of CIDR /28, subnet mask is 255.255.255.240
- Maximum number of users will be 13 hosts (minus the gateway IP address)

2nd Subnet:

- Network: 192.168.10.32/28
- Hosts: 192.168.10.33 - 192.168.10.46 (/28)
- Broadcast: 192.168.10.47/28

The screenshot shows the DHCP_SERVER configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'FastEthernet0' interface. The configuration includes a pool name 'HR_DEPT_DHCP', a default gateway of 192.168.10.33, and a subnet mask of 255.255.255.240. The maximum number of users is set to 13. The TFTP server and WLC address are both set to 0.0.0.0. A table at the bottom lists the configured DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
HR_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
IT_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
COMLAB_DHCP	192.168....	0.0.0.0	192.168....	255.255....	29	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

DHCP server configuration for VLAN 30 (IT Department):

- The first IP address will be allocated for default gateway (192.168.10.49)
- Given the subnet mask of CIDR /28, subnet mask is 255.255.255.240
- Maximum number of users will be 13 hosts (minus the gateway IP address)

3rd Subnet:

- Network: 192.168.10.48/28
- Hosts: 192.168.10.49 - 192.168.10.62 (/28)
- Broadcast: 192.168.10.63/28

DHCP_SERVER

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameIT_DEPT_DHCP

Default Gateway192.168.10.49

DNS Server0.0.0.0

Start IP Address : 1921681050

Subnet Mask: 255255255240

Maximum Number of Users :13

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
HR_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
IT_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
COMLAB_DHCP	192.168....	0.0.0.0	192.168....	255.255....	29	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

Top

DHCP server configuration for VLAN 40 (Helpdesk Room - guest wifi):

- The first IP address will be allocated for default gateway (192.168.10.65)
- Given the subnet mask of CIDR /28, subnet mask is 255.255.255.240
- Maximum number of users will be 13 hosts (minus the gateway IP address)

4th Subnet:

- Network: 192.168.10.64/28
- Hosts: 192.168.10.65 - 192.168.10.78 (/28)
- Broadcast: 192.168.10.79/28

The screenshot shows the DHCP_SERVER configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'FastEthernet0' interface. The configuration includes a pool name 'GUEST_WIFI_DHCP', a default gateway of 192.168.10.65, a DNS server of 0.0.0.0, a start IP address of 192.168.10.66, a subnet mask of 255.255.255.240, and a maximum number of users of 13. The TFTP and WLC addresses are set to 0.0.0.0. A table at the bottom lists the configured DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
GUEST_WIFI_D...	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
HR_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
IT_DEPT_DHCP	192.168....	0.0.0.0	192.168....	255.255....	13	0.0.0.0	0.0.0.0
COMLAB_DHCP	192.168....	0.0.0.0	192.168....	255.255....	29	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	512	0.0.0.0	0.0.0.0

DHCP server configuration for VLAN 50 (SERVER):

- The first IP address will be allocated for default gateway (192.168.10.81)
- Given the subnet mask of CIDR /29, subnet mask is 255.255.255.248
- Maximum number of users will be 5 hosts (minus the gateway IP address)

5th Subnet (Server):

- Network: 192.168.10.80/29
- Hosts: 192.168.10.81 - 192.168.10.86 (/29)
- Broadcast: 192.168.10.87/29

